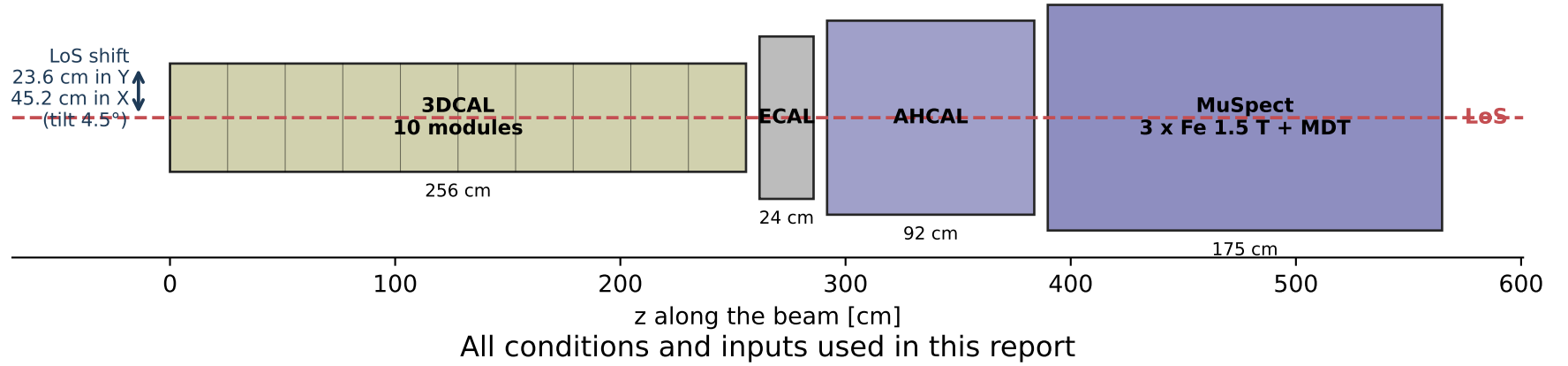


FASERCal layout, to scale (side view) — this study uses the 3DCAL only



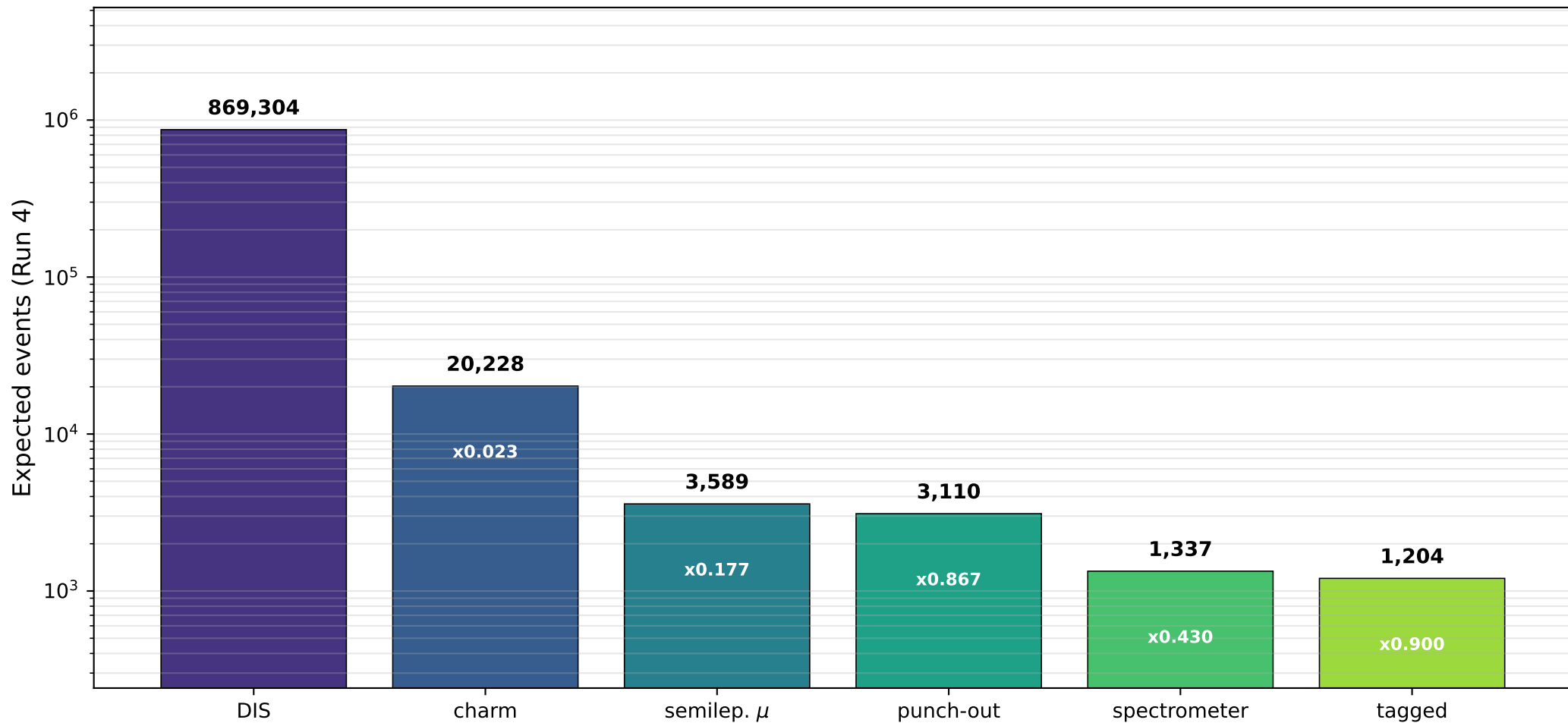
condition	value	source
Luminosity (Run 4 nominal)	780 fb^{-1}	FASERCal Bern CM talk
Fiducial volume	3DCAL only, AHCAL excluded	as requested
3DCAL geometry	10 modules x 20 layers of 1 cm^3 cubes	talk slide 4
3DCAL face	$48 \times 48 \text{ cm}$	talk slide 11
Absorber	1 mm or 5 mm W per MODULE (every 20 layers)	talk slide 39
3DCAL total mass	581 kg (1 mm) / 896 kg (5 mm)	CDR Table 5
Fiducial cut	none -- 3DCal volume only	CDR Table 8
Target mass	581 kg (1 mm) / 896 kg (5 mm)	derived
Charge misidentification	0.7% ($< 100 \text{ GeV}$) $\rightarrow$ 2-5% (TeV)	CDR Sec. 2.6.2
Spectrometer	10 Fe planes 1.5 T, 11 SciFi stations $100 \mu\text{m}$	CDR Table 7
Radiation / collision length (5 mm)	$18.3 X_0 / 4.8 \lambda$	CDR Table 5
Position	LoS shift (572, 236) mm, tilt 4.5°	talk slides 11-12
Target composition	polystyrene CH, $w_p=0.538$ (+ W where present)	computed per event
Muon flux	FASERnu Run 3 LHAPDF grid 'var2' ($25 \times 30 \text{ cm}$)	arXiv:2506.13889 Eq. 2.1
Off-axis flux factor	$F_{\text{flux}} = 1.10$ at (+452,+236) mm	FLUKA map, this work
Run 4 optics enhancement	$F_{\text{optics}} = 2.0$ (rate $\times 2$, harder)	CDR Sec. 1.3.2
AHCAL as extra target (optional)	3.33 t total, 1.60 t fiducial, $\sim 98\%$ Fe	CDR Table 3 / talk
Detector side	left of LoS looking downstream = +x	user; right-handed frame
Spectrometer acceptance	43% (≥ 2 MDT stations)	talk slide 33
Generator	POWHEG + Pythia8, NNPDF4.0 fitted vs perturbative charm	10 seeds, 764k events
Charge tag	semileptonic μ charge = charm sign	verified in code, 100%

Muon-DIS charm chain in FASERcal (AHCAL excluded)

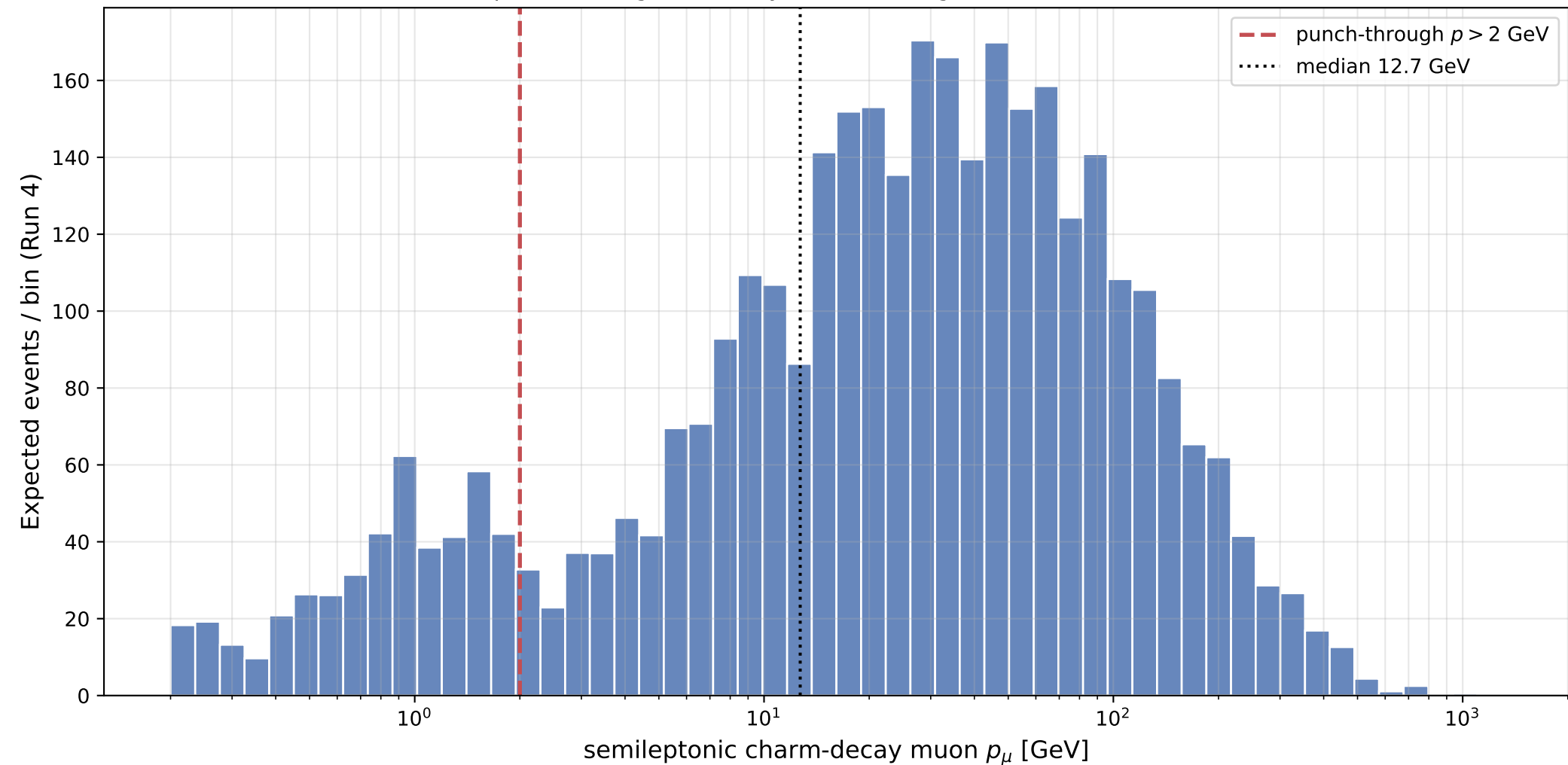
FASERcal chain — Run 4, 780 fb ⁻¹	3DCAL only (581 kg)	3DCAL + AHCAL (3908 kg)	fraction
Muon DIS interactions in fiducial volume	869,304	5,784,967	100%
... containing charm (≥ 1 charm hadron)	20,228	135,764	2.33%
... charm decaying semileptonically to μ	3,589	24,160	17.7% of charm
... μ punches out ($p > 2$ GeV)	3,110	20,886	86.7%
... reaches spectrometer (43%)	1,337	9,016	43%
... identified + linked ($\varepsilon=0.90$)	1,204	8,114	0.138% of DIS
Charge-tag purity (c vs $\bar{c}$)	99.3%	99.2%	CDR Sec. 2.6.2
Statistical reach on A_c	± 0.029	± 0.011	

3DCAL geometry and mass are now the DESIGNED values (Bern CM talk, 15 Jul 2026): 10 modules x 20 layers of 1 cm cubes, 48x48 cm face, W per module.
The detector sits LEFT of the LoS looking downstream, i.e. at (+452, +236) mm, where the FLUKA map gives $F_{\text{flux}} = 0.96$ — the quieter lobe.
(The mirror position would have given 2.02, with a harder, mu+-enriched beam.) Yields scale linearly with F_{flux} and with the efficiency; the chain fractions are ratios.

Chain attrition — 780 fb⁻¹, 0.6 t fiducial. Overall 0.138% of DIS events give a signed, linked charm muon

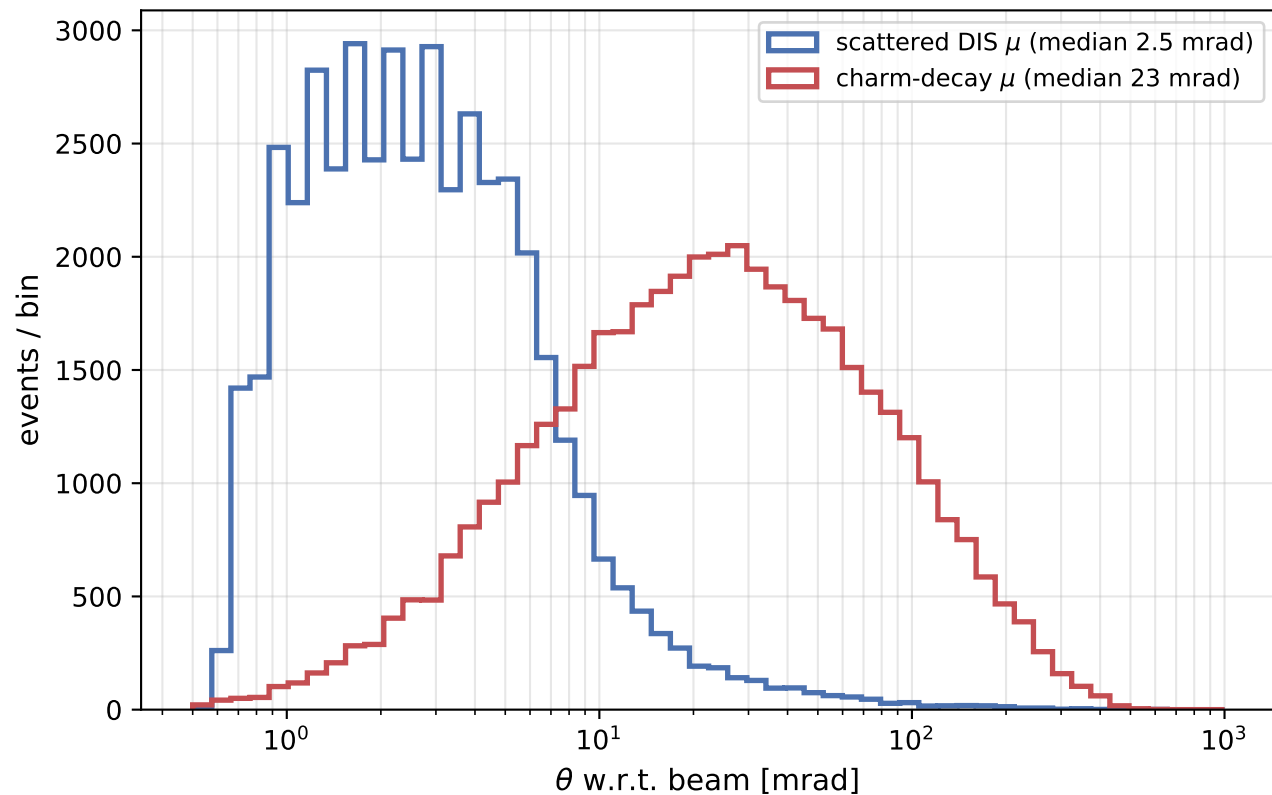


The muon that carries the charm charge: this spectrum sets both the punch-through efficiency and the charge-confusion dilution

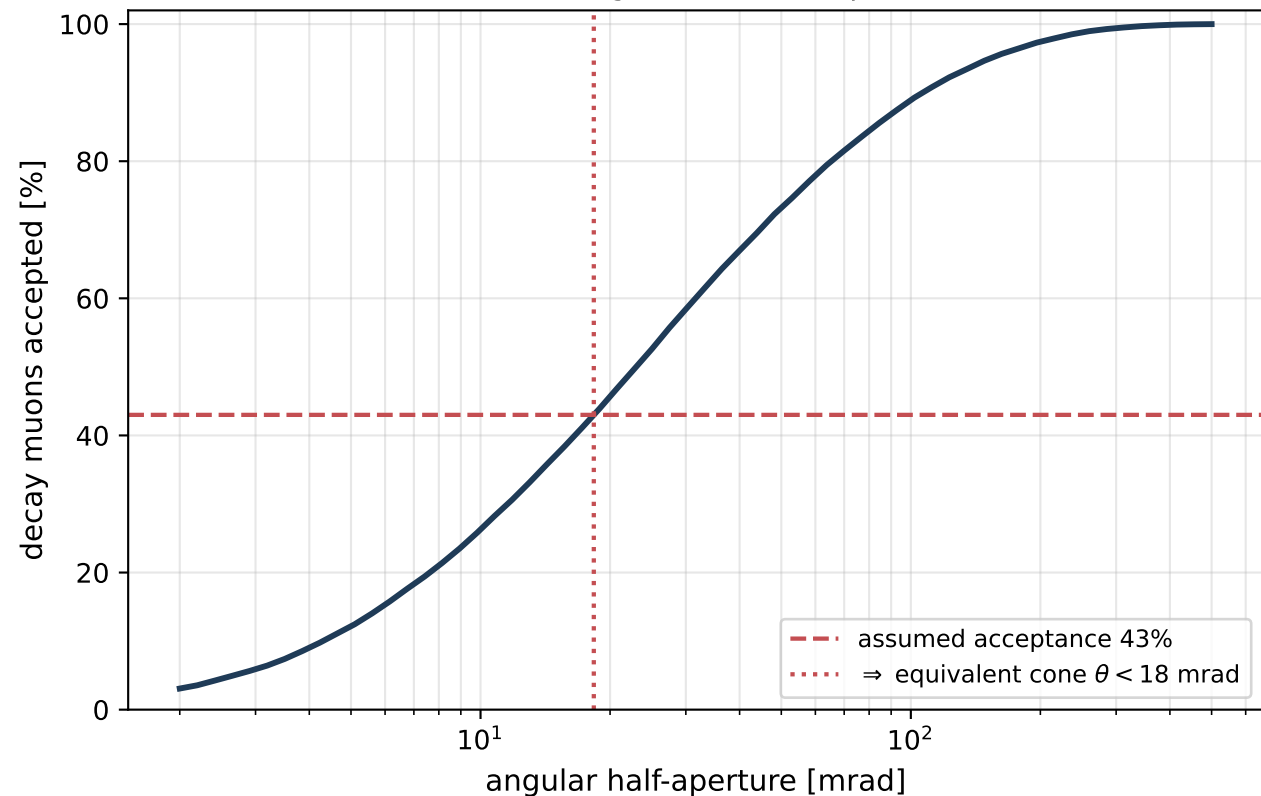


Acceptance is an ANGULAR problem, not a momentum one

The decay muon is an order of magnitude wider than the scattered muon

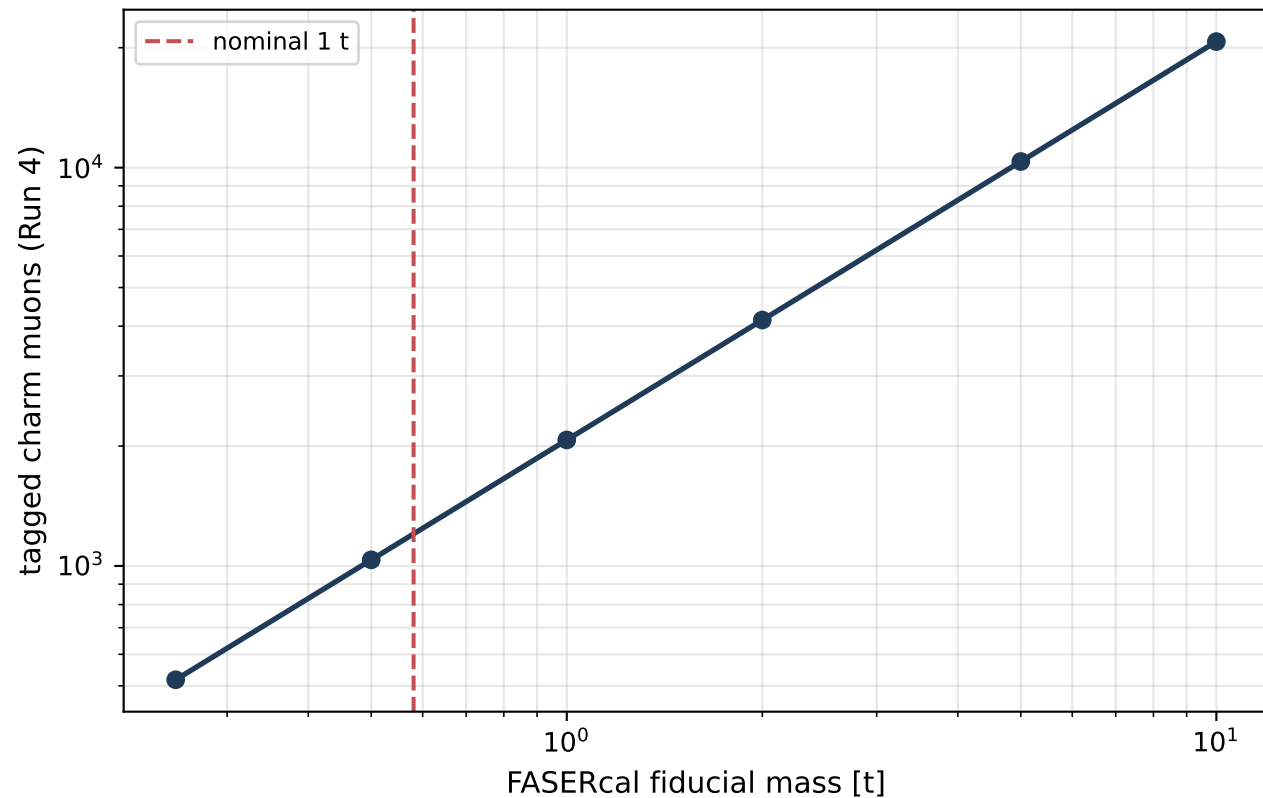


Assumed 40% acceptance $\equiv$ a $\theta < 18$ mrad cone
(checkable against the real aperture)

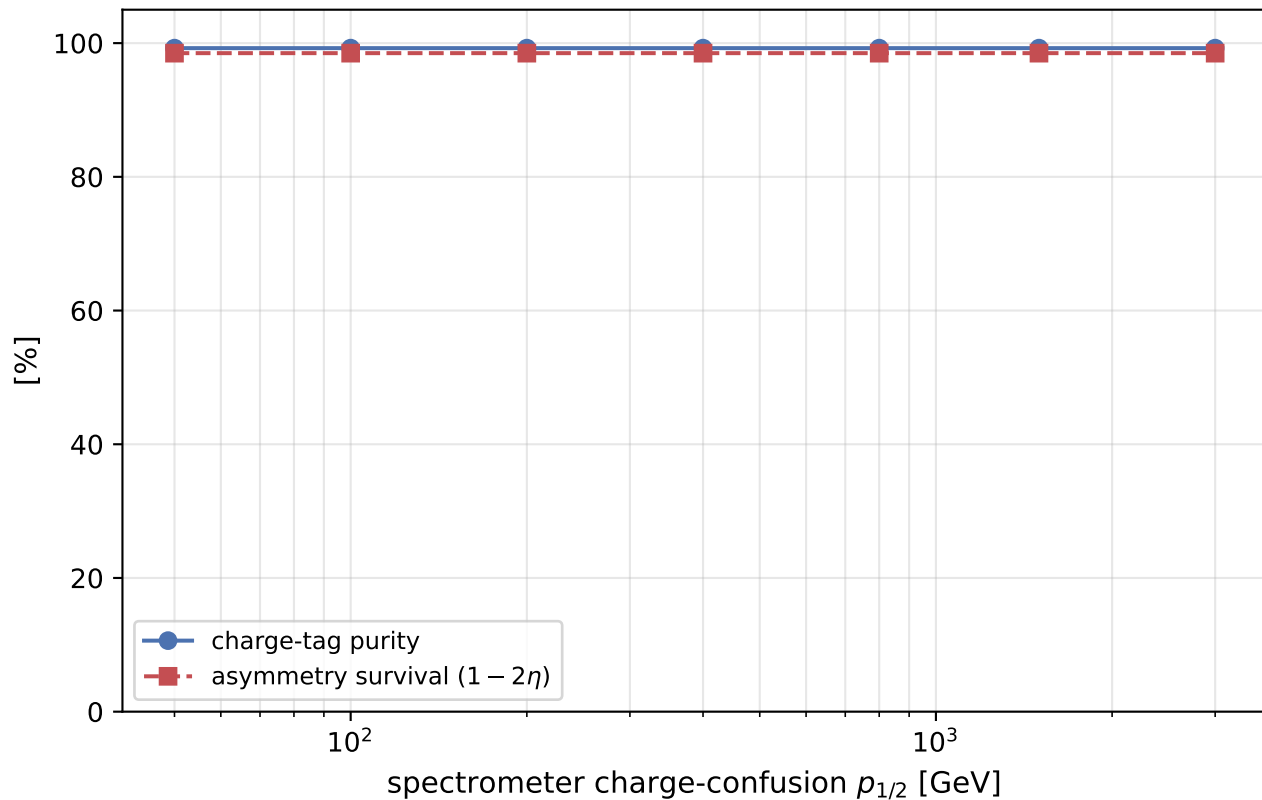


Dependence on the two dominant assumptions

Yield is LINEAR in fiducial mass
(rescale when 3DCAL geometry is fixed)

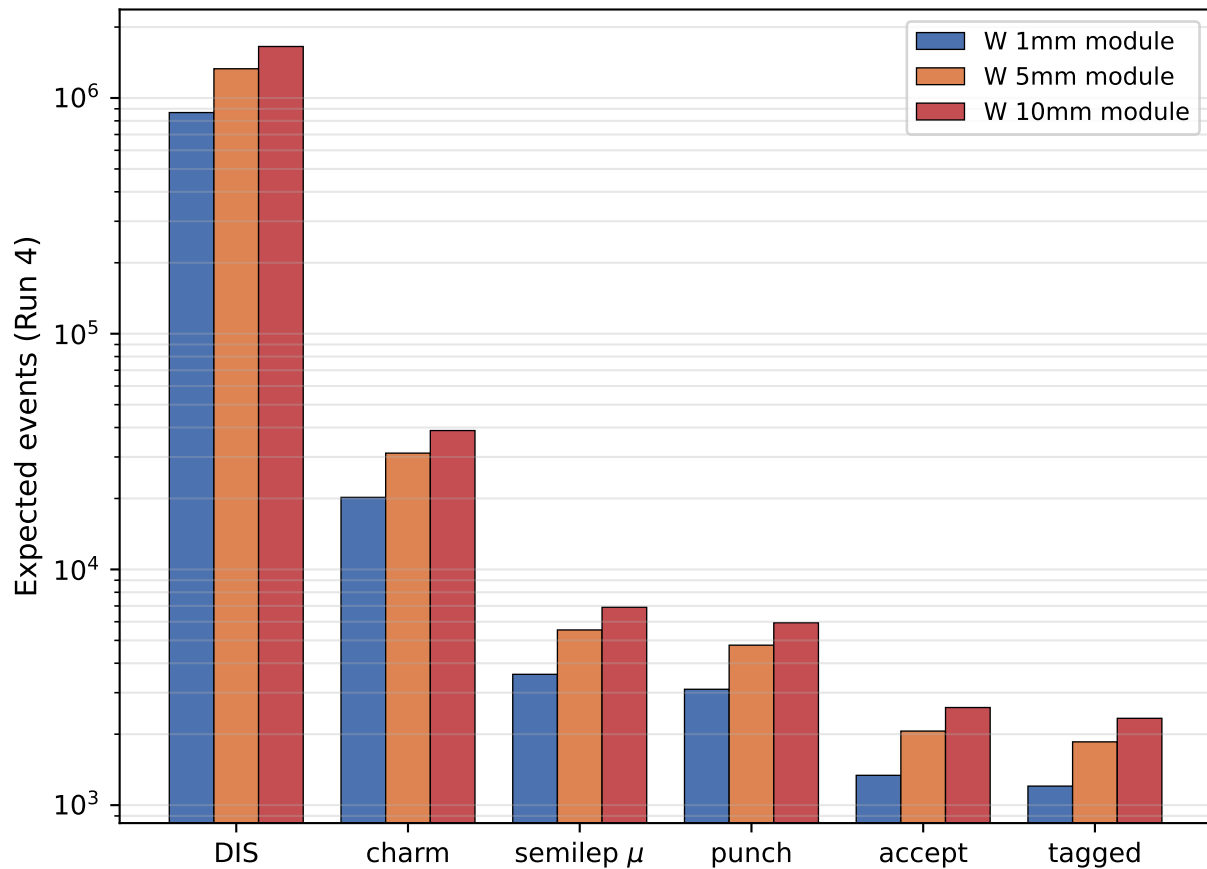


Charge ID is NOT the bottleneck:
the decay muons are soft in absolute terms

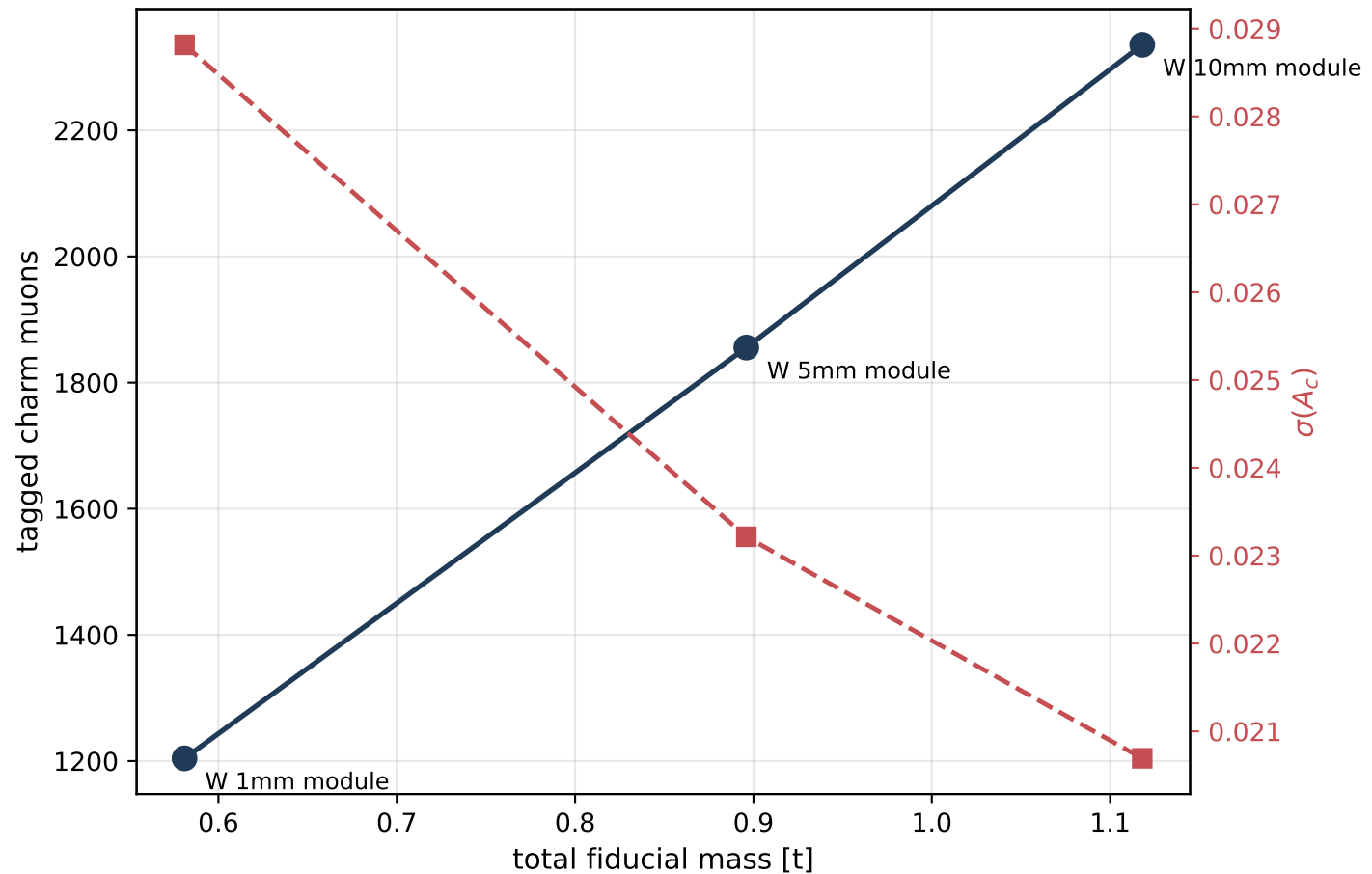


Tungsten absorber options as designed — 10 modules x 20 layers, W per module

Tungsten multiplies the whole chain



Yield tracks mass; $\sigma(A_c)$ improves as $1/\sqrt{N}$



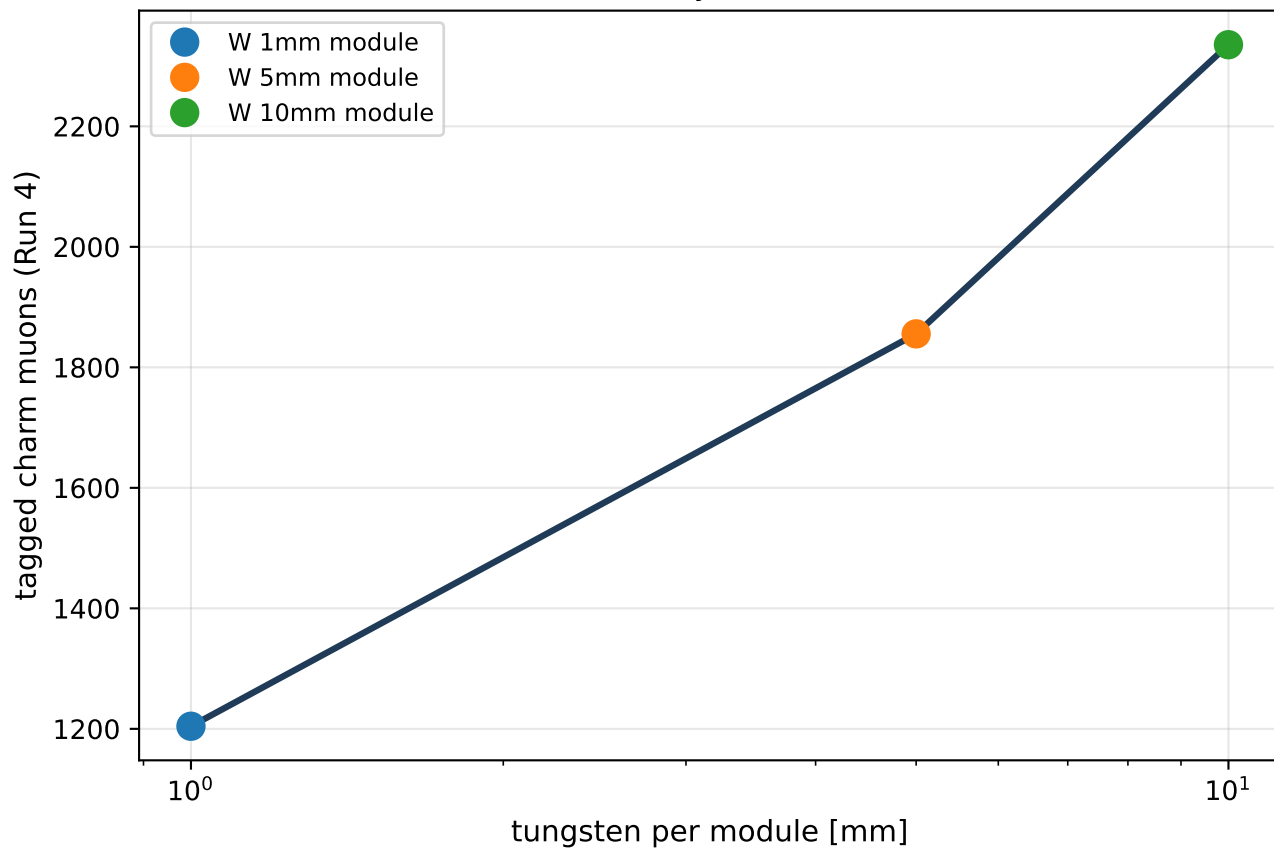
Tungsten scenarios — material budget and physics reach

scenario	M scint/W [t]	M tot [t]	$\langle X_0 \rangle$	θ_{MS} [mrad]	accept.	tagged	$\sigma(A_c)$	gain
W 1mm module	0.53 / 0.05	0.58	3.8	3.1	43.1%	1,204	0.029	x1.00
W 5mm module	0.61 / 0.29	0.90	9.6	5.1	43.2%	1,856	0.023	x1.24
W 10mm modul	0.57 / 0.54	1.12	16.7	6.9	43.7%	2,335	0.021	x1.39

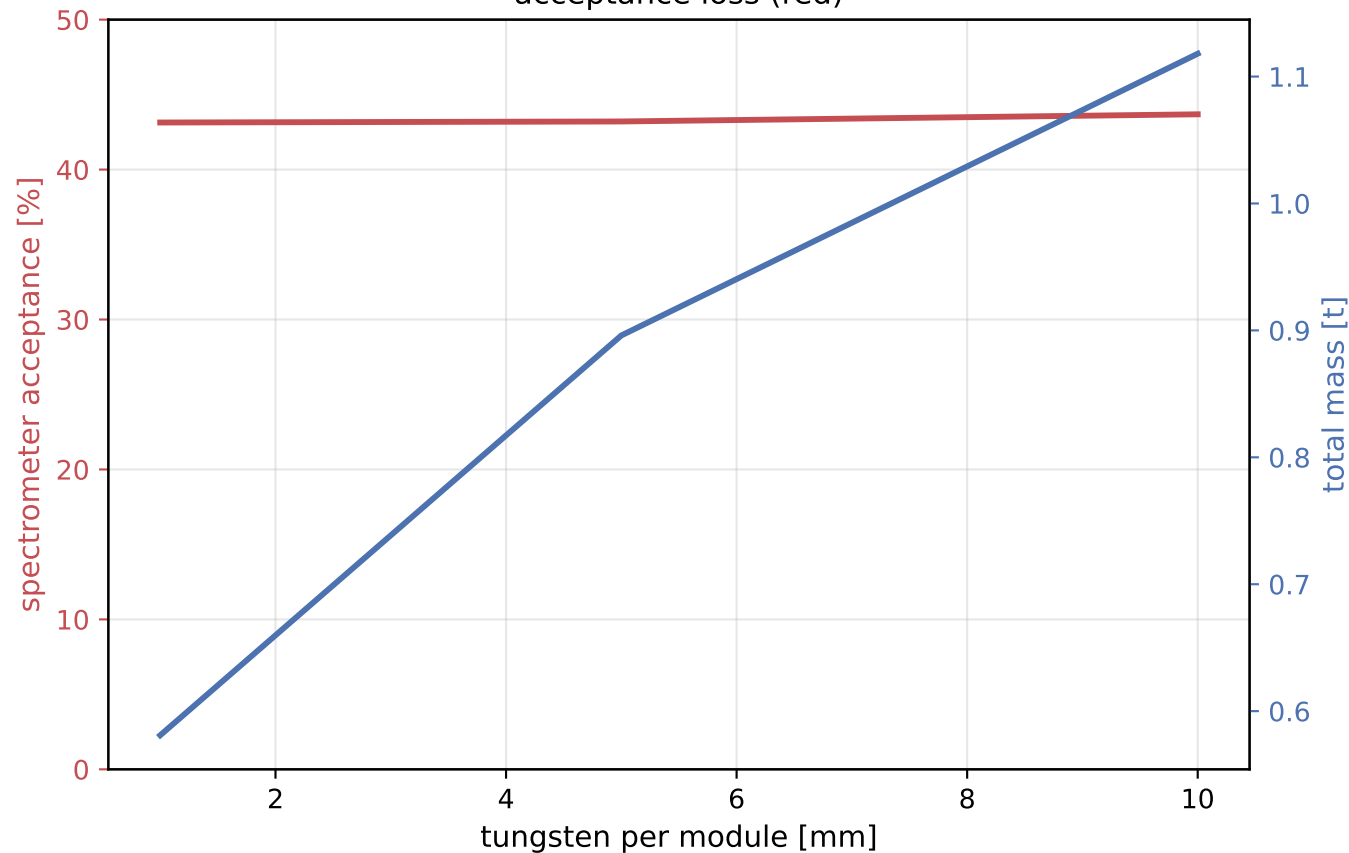
*Multiple scattering grows steeply (θ_{MS} 1.9 $\rightarrow$ 13.4 mrad) but costs only a few % of acceptance:
the decay-muon angular spread is intrinsically much wider than the aperture, so scattering moves nearly as many muons in as out.
The mass gain therefore wins outright.*

Absorber thickness per MODULE (every 20 layers, as designed) — yield rises monotonically; the limit is calorimetric

Yield rises monotonically with absorber thickness



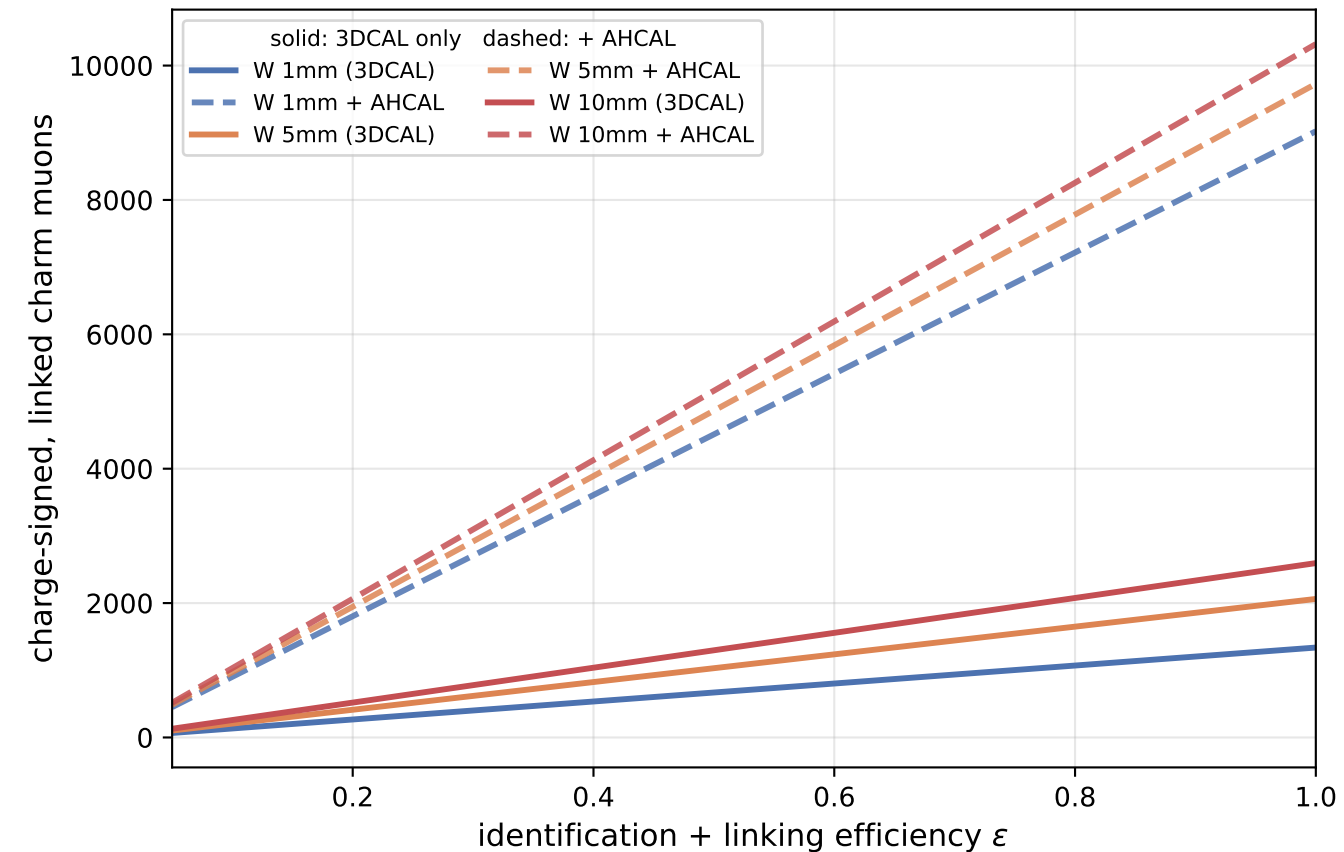
The trade-off: mass gain (blue) vs acceptance loss (red)



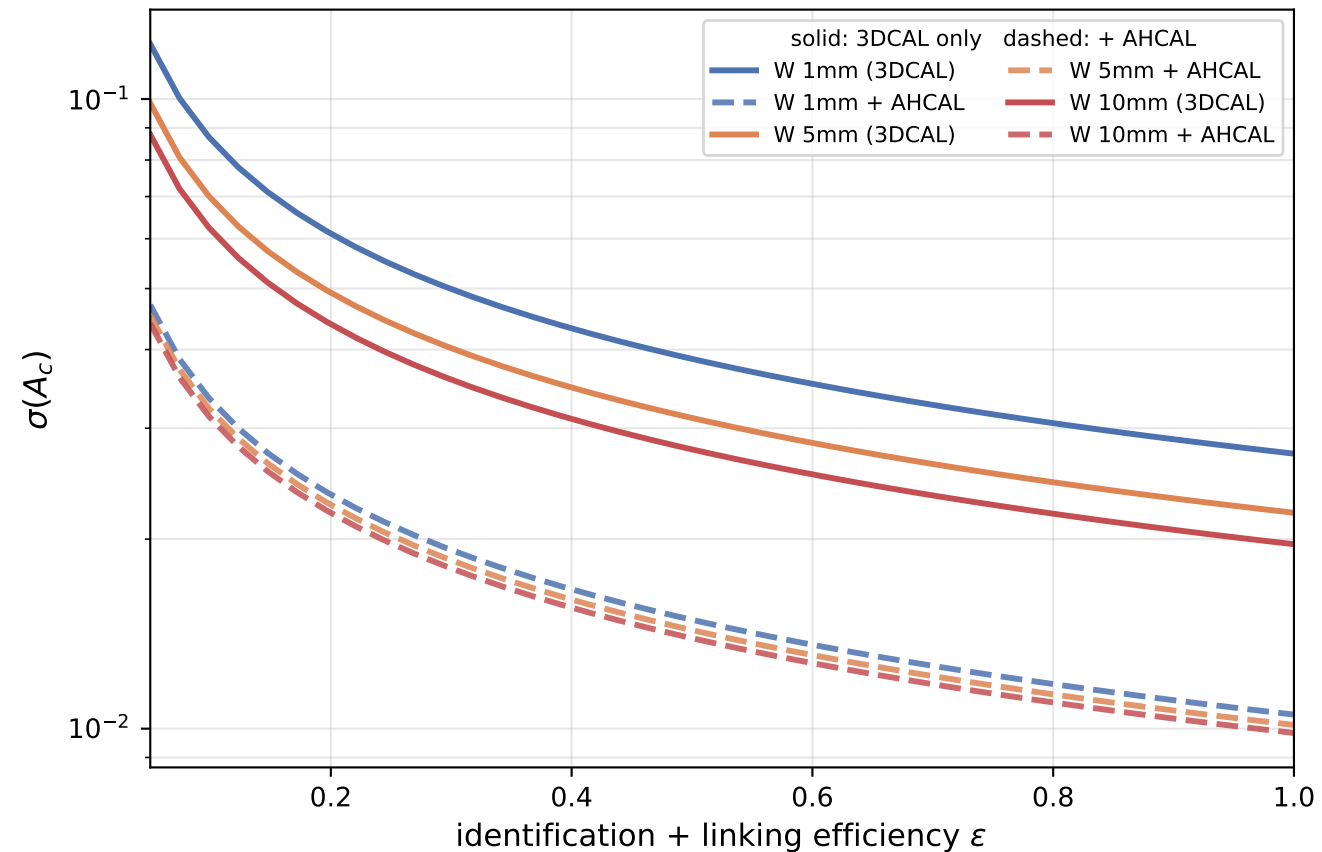
Main result vs the identification + linking efficiency (Run 4, 780 fb⁻¹, $F_{\text{flux}}=1.10$, $F_{\text{optics}}=2.0$)

The AHCAL adds $\sim 5.8\times$ the 3DCAL target mass, but its coarse 4x4 cm² granularity should be read as a LOWER ε — compare curves at different ε , not at the same one

Tagged yield is LINEAR in the efficiency

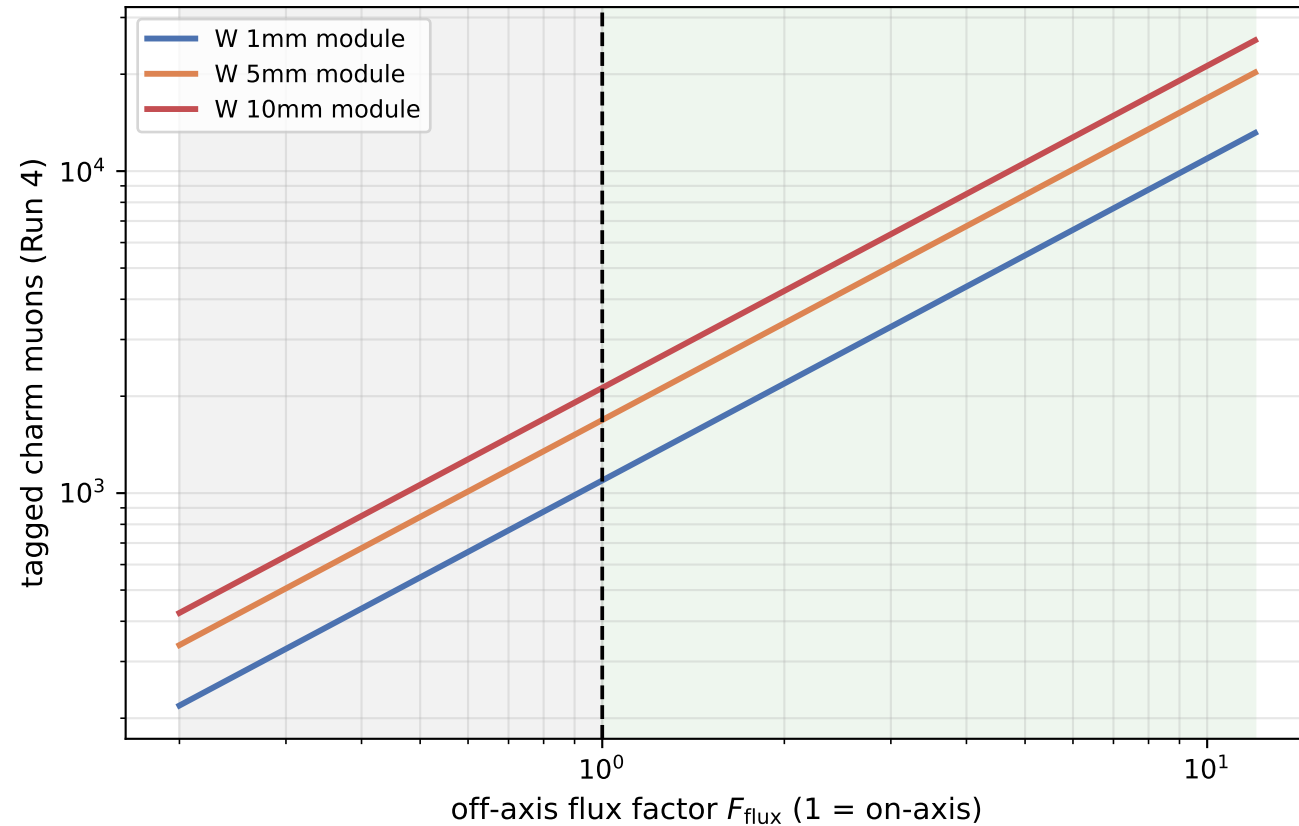


Statistical reach: $\sigma(A_c) \propto 1/\sqrt{\varepsilon}$

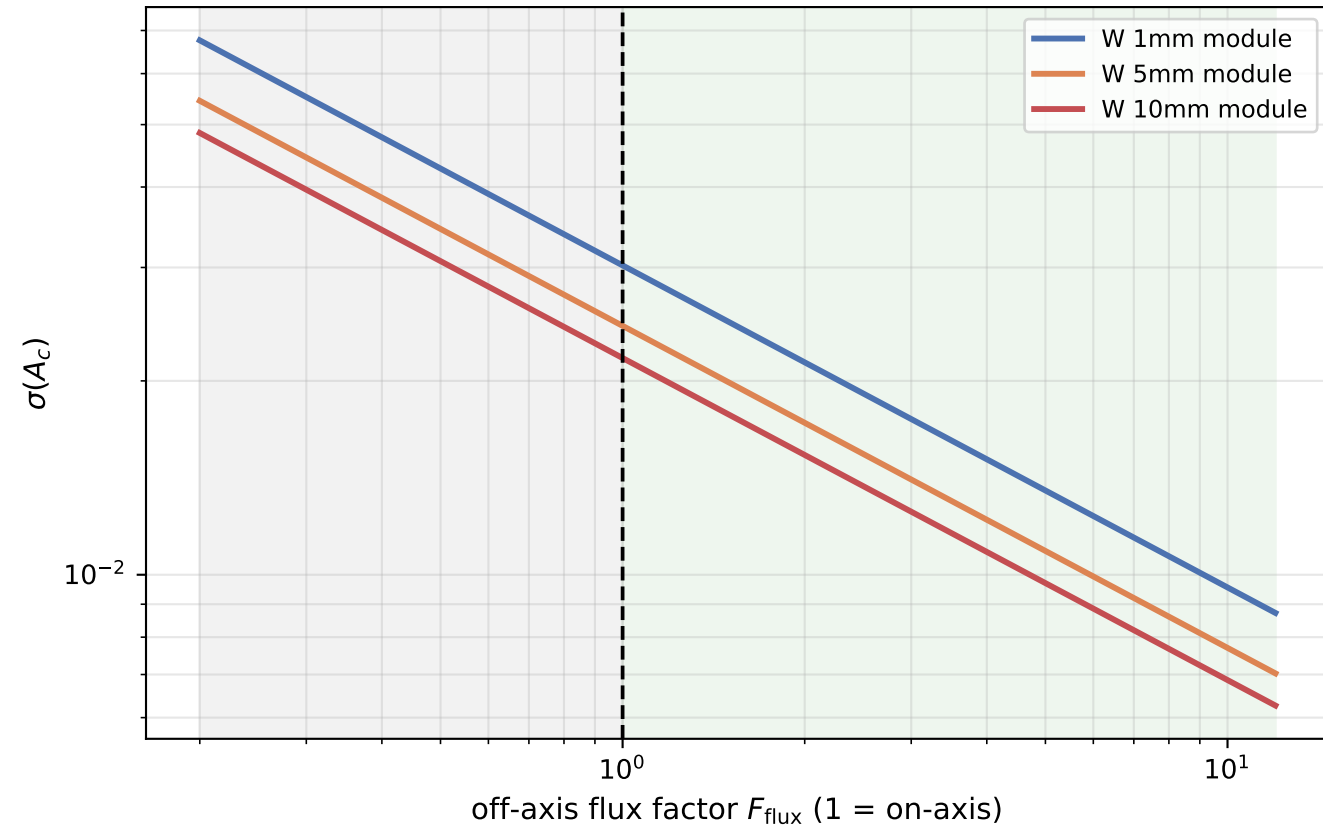


Off-axis flux — the largest unquantified factor. FLUKA reports the rate RISING off-axis in some directions (green), so this is not necessarily a penalty. REQUIRED INPUT: fluence at the 3DCAL position.

Yield scales linearly with the off-axis flux



$$\sigma(A_c) \propto 1/\sqrt{F_{\text{flux}}}$$



Intrinsic-charm sensitivity: the two PDF hypotheses through the same chain
(inclusive ratios; the IC signal lives at large x, not in the total)

